

According to the New Book & Syllabus of  
Board of Intermediate, Sindh.

# THE STUDENTS' COMPANION OF *Mathematics*

For Class **11**

INCLUDES **MODEL PAPER** FOR FURTHER PRACTICE





## Chapter # 06 PERMUTATION, COMBINATION & PROBABILITY

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## Chapter # 07 MATHEMATICAL INDUCTION & BINOMIAL THEOREM

# CHAPTER # 06 PERMUTATION, COMBINATION & PROBABILITY

## EXERCISE 6.1

Q.1 Evaluate the following:

$$(i) 4! \quad (ii) 6! \quad (iii) \frac{8!}{5!} \quad (iv) \frac{10!}{7!} \quad (v) 5! \times 7! \times 3! \quad (vi) 10! \div 8! + 5!$$

$$(vii) \frac{16!}{8!12!}$$

$$(viii) \frac{(9!)^2}{(5!)^3 \times 7!}$$

$$(ix) \frac{9!}{2! \times (9-2)!}$$

**SOLUTION:**

$$(i) 4! = 4(3)(2)(1) = 24$$

$$(ii) 6! = 6(5)(4)(3)(2)(1) = 720$$

$$(iii) \frac{8!}{5!} = \frac{8(7)(6)(5!)}{5!} = 8(7)(6) = 336$$

$$(iv) \frac{10!}{7!} = \frac{10(9)(8)(7!)}{7!} = 10(9)(8) = 720$$

$$(v) 5! \times 7! \times 3! = 120 \times 5040 \times 6 = 3628800$$

$$(vi) 10! \div 8! + 5! = \frac{10!}{8!} + 120 = \frac{10(9)(8!)}{8!} + 120 = 10(9) + 120 \\ = 90 + 120 = 210$$

$$(vii) \frac{16!}{8!12!} = \frac{16(15)(14)(13)(12!)}{8(7)(6)(5)(4)(3)(2)(1)12!} = \frac{16(15)(14)(13)}{8(7)(6)(5)(4)(3)(2)(1)} = \frac{13}{12}$$

$$(viii) \frac{(9!)^2}{(5!)^3 \times 7!} = \frac{\{(9)(8)(7)(6)(5!)\}^2}{(5!)^3 \times (7)(6)(5)(4)(3)(2)(1)} = \frac{\{(9)(8)(7)(6)\}^2 (5!)^2}{(5!)^3 \times (5040)} \\ = \frac{(3024)^2}{(120) \times (5040)} = \frac{378}{25}$$

$$(ix) \frac{9!}{2! \times (9-2)!} = \frac{9(8)(7!)}{2 \times (7)!} = \frac{9(8)}{2!} = 36$$

Q.2 Write the following in factorial form:

$$(i) 6 \cdot 5 \cdot 4 \quad (ii) 12 \cdot 11 \cdot 10$$

$$(iii) \frac{n(n+1)(n+2)}{6 \cdot 5}$$

$$(iv) 20 \cdot 19 \cdot 18 \cdot 17$$

**SOLUTION:**

$$(i) 6 \cdot 5 \cdot 4$$



Multiply and divide by 3!

$$= \frac{6 \cdot 5 \cdot 4 \cdot 3!}{3!} = \frac{6!}{3!}$$

(ii)  $12 \cdot 11 \cdot 10$

Multiply and divide by 9!

$$= \frac{12 \cdot 11 \cdot 10(9!)}{9!} = \frac{12!}{9!}$$

(iii)  $\frac{n(n+1)(n+2)}{6 \cdot 5}$

Multiply and divide by  $4!$  and  $(n-1)!$

$$= \frac{n(n+2)(n+1)4!(n-1)!}{6 \cdot 5 \cdot 4!(n-1)!}$$

$$= \frac{(4!)(n+2)!}{(6!)(n-1)!}$$

$$(iv) 20 \cdot 19 \cdot 18 \cdot 17 = \frac{20 \cdot 19 \cdot 18 \cdot 17(16!)}{16!} = \frac{20!}{16!}$$

Q.3 Express the following as a single fraction:

$$(i) \frac{(n+1)!}{(r+1)!} + \frac{n!}{r!}$$

$$(ii) \frac{(n+1)!}{r!} + \frac{n!}{(r+1)!}$$

**SOLUTION:**

$$(i) \frac{(n+1)!}{(r+1)!} + \frac{n!}{r!} = \frac{n!}{r!} \left[ \frac{n+1}{r+1} + 1 \right] = \frac{n!}{r!} \left[ \frac{n+1+r+1}{r+1} \right] = \frac{n!(n+r+2)}{(r+1)!}$$

$$(ii) \frac{(n+1)!}{r!} + \frac{n!}{(r+1)!} = \frac{(n+1)!}{r!} + \frac{n!}{(r+1)r!} = \frac{(n+1)!(r+1) + n!}{(r+1)r!}$$

$$= \frac{n![(n+1)(r+1)+1]}{(r+1)r!} = \frac{n![(nr+n+r+1)+1]}{(r+1)!} = \frac{n![nr+n+r+2]}{(r+1)!}$$

## EXERCISE 6.2

Q.1 Evaluate the following:



$$(i) {}^{20}P_3 (ii) {}^{16}P_4 (iii) {}^{12}P_5 (iv) \binom{8}{3,2} (v) \binom{9}{2,3,4}$$

**SOLUTION:**

$$(i) {}^{20}P_3 = \frac{20!}{(20-3)!} = \frac{20!}{17!} = 6840$$

$$(ii) {}^{16}P_4 = \frac{16!}{(16-4)!} = \frac{16!}{12!} = 43680$$

$$(iii) {}^{12}P_5 = \frac{12!}{(12-5)!} = \frac{12!}{7!} = 95040$$

$$(iv) \binom{8}{3,2} = \frac{8!}{3! \times 2!} = 3360$$

$$(v) \binom{9}{2,3,4} = \frac{9!}{2! \times 3! \times 4!} = 1260$$

**Q.2 Find the value of  $n$  in each of the following:**

$$(i) {}^nP_2 = 30$$

$$(ii) {}^nP_3 = 60$$

$$(iii) {}^8P_n = 8 \cdot 7 \cdot 6$$

$$(iv) {}^{11}P_n = 11 \cdot 10 \cdot 9$$

$$(v) {}^{n+2}P_4 : {}^{n+1}P_3 = 5 : 1$$

$$(vi) {}^nP_4 : {}^{n-1}P_3 = 9 : 1$$

**SOLUTION:**

$$(i) {}^nP_2 = 30 \Rightarrow \frac{n!}{(n-2)!} = 30 \Rightarrow \frac{n(n-1)(n-2)!}{(n-2)!} = 30$$

$$n(n-1) = 30 \Rightarrow n^2 - n - 30 = 0 \Rightarrow n^2 - 6n + 5n - 30 = 0$$

$$n(n-6) + 5(n-6) = 0 \Rightarrow (n-6)(n+5) = 0$$

$$n+5 = 0 \Rightarrow n = -5 \notin \mathbb{N}$$

$$n-6 = 0 \Rightarrow \boxed{n = 6}$$

$$(ii) {}^nP_3 = 60 \Rightarrow \frac{n!}{(n-3)!} = 60 \Rightarrow \frac{n(n-1)(n-2)(n-3)!}{(n-3)!} = 60$$

$$n(n-1)(n-2) = 60 \Rightarrow n(n^2 - 3n + 2) - 60 = 0$$

$$n^3 - 3n^2 + 2n - 60 = 0$$

$$n^3 - 5n^2 + 2n^2 - 10n + 12n - 60 = 0$$

$$n^2(n-5) + 2n(n-5) + 12(n-5) = 0$$

$$(n-5)(n^2+2n+12)=0$$

$$n-5 \Rightarrow \boxed{n=5}$$

$$n^2+2n+12=0$$

$$n = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(12)}}{2(1)}$$

$$n = \frac{-2 \pm \sqrt{4-48}}{2}$$

$$n = \frac{-2 \pm \sqrt{-44}}{2}$$

$$n = \frac{-2 \pm \sqrt{2*2*(-11)}}{2}$$

$$n = \frac{-2 \pm 2\sqrt{-11}}{2}$$

$$n = \frac{2(-1 \pm \sqrt{-11})}{2}$$

$$\boxed{n = -1 \pm \sqrt{-11} \notin \mathbb{N}}$$

$$(iii) {}^8P_n = 8.7.6 \Rightarrow \frac{8!}{(8-n)!} = 336 \Rightarrow \frac{8!}{336} = (8-n)!$$

$$120 = (8-n)! \Rightarrow (8-n)! = 5! \Rightarrow 8-n = 5 \Rightarrow 8-5 = n$$

$$\boxed{n=3}$$

$$(iv) {}^{11}P_n = 11 \cdot 10 \cdot 9 \Rightarrow \frac{11!}{(11-n)!} = 990 \Rightarrow \frac{11!}{990} = (11-n)!$$

$$40320 = (11-n)! \Rightarrow (11-n)! = 8! \Rightarrow 11-n = 8$$

$$11-8 = n \Rightarrow \boxed{n=3}$$

$$(v) {}^{n+2}P_4 : {}^{n+1}P_3 = 5:1 \Rightarrow \left\{ \frac{(n+2)!}{(n+2-4)!} \right\} \div \left\{ \frac{(n+1)!}{(n+1-3)!} \right\} = \frac{5}{1}$$



$$\left\{ \frac{(n+2)(n+1)n(n-1)(n-2)!}{(n-2)!} \right\} \div \left\{ \frac{(n+1)n(n-1)(n-2)!}{(n-2)!} \right\} = 5$$

$$\{n(n+2)(n+1)(n-1)\} \div \{n(n+1)(n-1)\} = 5$$

$$\frac{n(n+2)(n+1)(n-1)}{n(n+1)(n-1)} = 5$$

$$n+2 = 5 \Rightarrow n = 5-2 \Rightarrow \boxed{n=3}$$

(vi)  ${}^n P_4 : {}^{n-1} P_3 = 9 : 1$

$$\left\{ \frac{n!}{(n-4)!} \right\} \div \left\{ \frac{(n-1)!}{(n-1-3)!} \right\} = \frac{9}{1}$$

$$\left\{ \frac{n(n-1)(n-2)(n-3)(n-4)!}{(n-4)!} \right\} \div \left\{ \frac{(n-1)(n-2)(n-3)(n-4)!}{(n-4)!} \right\} = 9$$

$$\{n(n-1)(n-2)(n-3)\} \div \{(n-1)(n-2)(n-3)\} = 9$$

$$\frac{n(n-1)(n-2)(n-3)}{(n-1)(n-2)(n-3)} = 9 \Rightarrow \boxed{n=9}$$

**Q.3** In how many ways different batting orders are possible for a cricket team consisting of 11 players.

**SOLUTION:**

Total players = 11

Total ways = 11(10)(9)(8)(7)(6)(5)(4)(3)(2)(1)

Total ways = 39916800

**Q.4** In how many ways can two English books, three Chemistry books and four Physics books be arranged on a shelf so that all the books of same subject are together?

**SOLUTION:**

Total English books = 2

Total Chemistry books = 3

Total Physics books = 4

Total languages = 3

Total ways =  $(2!)(3!)(4!)(3!)$

Total ways = 1728

**Q.5** How many different arrangements be formed of the word 'MATRIX' using all letters? Also find the number of arrangements if each of them start with R.

**SOLUTION:**

Total words =  $6! = 720$

If R is at first place then remaining letters =  $6 - 1 = 5$

Total words =  $5! = 120$

**Q.6** How many signals can be given by 8 flags of different colours when 5 of them are used at a time?

**SOLUTION:**

Total flags = 8

Total flags to be used = 5

Total signals =  ${}^8P_5 = 6720$

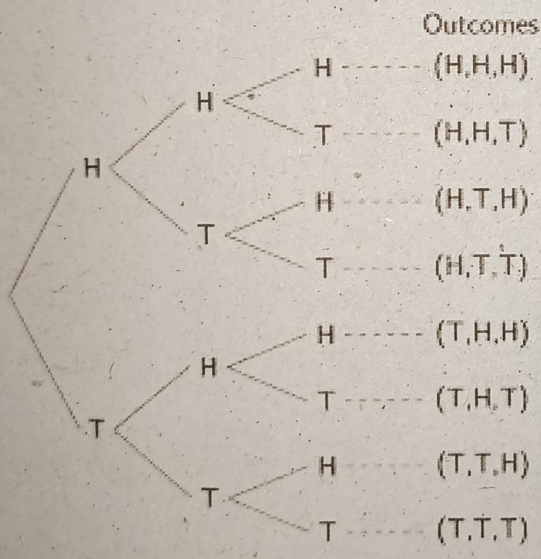
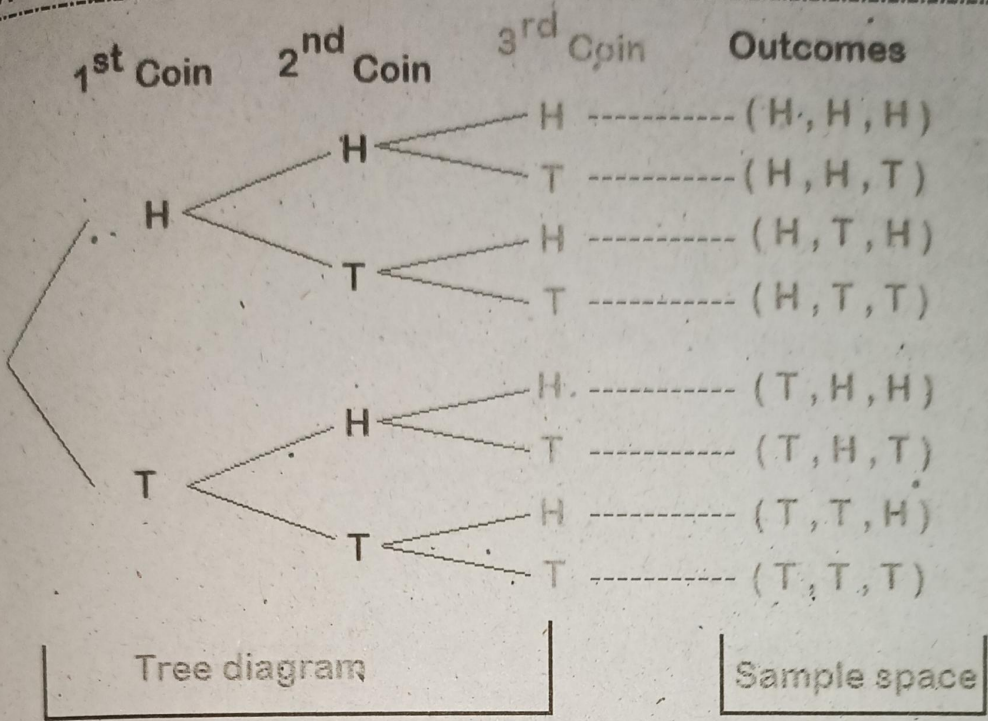
**Q.7A** A fair coin is tossed three times. How many outcomes are possible? Find by fundamental principle and illustrate using tree diagram.

**SOLUTION:**

Total coins = 3

Total outcomes =  $(2)(2)(2) = 8$





Q.8 How many 5-digit number can be formed, without repeating any digit:

- (i) 5,6,7,8,9      (ii) 1,2,3,5,7,9      (iii) 2,4,6,8,0

**SOLUTION:**

- (i) 5, 6, 7, 8, 9

Total numbers =  ${}^5P_5 = 120$

(ii) 1, 2, 3, 5, 7, 9

$$\text{Total numbers} = {}^6P_5 = 720$$

(iii) 2, 4, 6, 8, 0

$$\text{Total numbers} = {}^5P_5 - (1) {}^4P_4 = 96$$

Q.9 Prove that: (i)  ${}^nP_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

(ii)  ${}^nP_r = n \cdot {}^{n-1}P_{r-1}$

SOLUTION:

$$(i) \quad {}^nP_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$$

$$\text{R.H.S} = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$$

$$\text{R.H.S} = \frac{(n-1)!}{(n-1-r)!} + \frac{r(n-1)!}{\{n-1-(r-1)\}!}$$

$$\text{R.H.S} = \frac{(n-1)!}{(n-1-r)!} + \frac{r(n-1)!}{(n-1-r+1)!}$$

$$\text{R.H.S} = \frac{(n-1)!}{(n-1-r)!} + \frac{r(n-1)!}{(n-r)!}$$

$$\text{R.H.S} = \frac{(n-1)!}{(n-r-1)!} + \frac{r(n-1)!}{(n-r)(n-r-1)!}$$

$$\text{R.H.S} = \frac{(n-1)!}{(n-r-1)!} \left\{ 1 + \frac{r}{n-r} \right\}$$

$$\text{R.H.S} = \frac{(n-1)!}{(n-r-1)!} \left\{ \frac{n-r+r}{n-r} \right\}$$

$$\text{R.H.S} = \frac{(n-1)!}{(n-r-1)!} \left\{ \frac{n}{n-r} \right\} = \frac{n(n-1)!}{(n-r)(n-r-1)!}$$

$$\text{R.H.S} = \frac{n!}{(n-r)!}$$

$$\text{R.H.S} = {}^nP_r$$



$$(ii) {}^n P_r = n \cdot {}^{n-1} P_{r-1}$$

$$R.H.S = n \cdot {}^{n-1} P_{r-1}$$

$$R.H.S = \frac{n(n-1)!}{\{n-1-(r-1)\}!}$$

$$R.H.S = \frac{n!}{(n-1-r+1)!}$$

$$R.H.S = \frac{n!}{(n-r)!}$$

$$R.H.S = {}^n P_r$$

**Q.10** How many arrangements can be formed using all the letters of the following words.

(i) STATISTICS

(ii) PLANE

(iii) OBJECT

(iv) FASTENS

(v) MATHEMATICS

(vi) ASSISTANCE

**SOLUTION:**

**(i) STATISTICS**

Total letters = 10

$S = 3, T = 3, I = 2$

$$\text{Total words} = \frac{10!}{3!3!2!}$$

Total words = 50400

**(ii) PLANE**

Total letters = 5

Total words =  $5!$

Total words = 120

**(iii) OBJECT**

Total letters = 6

Total words = 6!

Total words = 720

#### (iv) FASTENS

Total letters = 7

S = 2

Total words =  $\frac{7!}{2!}$

Total words = 2520

#### (v) MATHEMATICS

Total letters = 11

M = 2, T = 2, A = 2

Total words =  $\frac{11!}{2! 2! 2!}$

Total words = 4989600

#### (vi) ASSISTANCE

Total letters = 10

A = 2, S = 3

Total words =  $\frac{10!}{2! 3!}$

Total words = 302400

**Q.11** How many 8-digit numbers can be formed from the digits 2, 2, 2, 3, 3, 3, 5, 6? How many among these are greater than 60,000,000?

**SOLUTION:**



$$(i) \text{ Total 8 digit numbers} = \frac{8!}{3!3!}$$

$$\text{Total 8 digit numbers} = 1120$$

$$(ii) \text{ Total 8 digit numbers} = \frac{(1)(7!)}{3!3!}$$

$$\text{Total 8 digit numbers} = 140$$

**Q.12** How many distinct permutations of letters of the word "ESSENTIAL" are possible? How many will have the two S's

(i) together (ii) separate?

**SOLUTION:**

$$\text{Total letters} = 8$$

$$E \underline{S} = 2$$

$$(i) \text{ Total words} = \frac{8!}{2!}$$

$$\text{Total words} = 20160$$

$$(ii) \text{ Total words} = \frac{9!}{2!2!} - \frac{8!}{2!}$$

$$\text{Total words} = 70560$$

**Q.13** In how many ways, can 8 persons be seated around a round table?

**SOLUTION:**

$$\text{Total arrangements in a circle} = (n - 1)!$$

$$\text{Total arrangements} = (8 - 1)!$$

$$= 7!$$

$$= 5040$$

**Q.14** In how many ways, 7 keys be arranged in a circular key ring?

**SOLUTION:**

A Help Book 31  
Total arrangements in a key ring =  $\frac{(n-1)!}{2}$

$$\text{Total arrangements} = \frac{(7-1)!}{2}$$

$$= \frac{6!}{2}$$

$$= 360$$

Q.15 How many 3-digit number can be formed by using each one of the digit 2, 3, 5, 7, 9 only once?

SOLUTION:

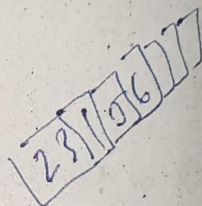
Total digits = 5

Digits may not be repeated

$$\text{Total 3 digit numbers} = (5)(4)(3) = 60$$

$$n = 5, r = 3$$

$${}^5P_3$$



Q.16 How many numbers greater than 23,000 can be formed from the digits 1, 2, 3, 5, 6 if digits may repeat?

SOLUTION:

$$\begin{aligned} \text{Total number} &= (1)(3)(5)(5)(5) + (3)(5)(5)(5)(5) \\ &= 375 + 1875 = 2250 \end{aligned}$$

Q.17 Find the number of 5-digit numbers that can be formed from the digits 1, 2, 4, 6, 8 (when no digit is repeated), but

(i) the digits 2 and 8 are next to each other

(ii) the digits 2 and 8 are not next to each other

SOLUTION:

$$1 \ 2 \ 8$$

$$(i) \text{ Total number} = (4!)(2!) = 48$$

$$(ii) \text{ Total number} = 5! - (4!)(2!) = 72$$

Q.18 How many 6-digit numbers can be formed without repeating any digit from the digits 0, 1, 2, 3, 4, 5? In how many of them will 0 be at the tenth place?

$$(324)$$

$$5 \times 5 \times 4 \times 3 \times 2 \times 1 = 600$$

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 0 | 2 | 3 | 4 | 5 |
| 2 | 2 | 3 | 4 | 5 |   |
| 3 | 3 | 4 | 5 |   |   |
| 4 | 4 | 5 |   |   |   |



**SOLUTION:**

(i) 0 cannot be at first place

$$\text{Total number} = (5)(5!) = 600$$

(ii) 0 be at tenth place

$$\text{Total number} = 5! (1) = 120$$

Q.19 How many 5-digit multiplies of 5 can be formed from the digits 2,3,5,7,9 when digits may repeat?

**SOLUTION:**

There are 5 options for each place except unit place

$$\text{Total number} = (5)(5)(5)(5)(1) = 625$$

**EXERCISE 6.3**

Q.1 Evaluate the following:

(i)  ${}^{12}C_{10}$

(ii)  ${}^{12}C_3$

(iii)  $\binom{20}{17}$

(iv)  $\binom{7}{4}$

**SOLUTION:**

$$(i) {}^{12}C_{10} = \frac{12!}{10!(12-10)!} = \frac{12!}{10!2!} = 66$$

$$(ii) {}^{12}C_3 = \frac{12!}{3!(12-3)!} = \frac{12!}{3!9!} = 220$$

$$(iii) \binom{20}{17} = \frac{20!}{17!(20-17)!} = \frac{20!}{17!(20-17)!} = 1140$$

$$(iv) \binom{7}{4} = \frac{7!}{4!(7-4)!} = \frac{7!}{4!(3)!} = 35$$

Q.2 Find the value of  $n$  when:

(i)  ${}^nC_2 = 21$

(ii)  ${}^nC_3 = 4$

(iii)  ${}^nC_{12} = {}^nC_6$

(iv)  ${}^nC_{15} = \frac{17 \times 16}{2!}$

(v)  ${}^nC_5 = {}^nC_4$

(vi)  ${}^nC_{10} = \frac{12 \times 11}{2!}$

**SOLUTION:**

$$(i) {}^nC_2 = 21 \Rightarrow \frac{n!}{2!(n-2)!} = 21 \Rightarrow \frac{n(n-1)(n-2)!}{2!(n-2)!} = 21$$

$$\frac{n(n-1)}{2} = 21 \Rightarrow n(n-1) = 42 \Rightarrow n^2 - n = 42$$

$$n^2 - n - 42 = 0 \Rightarrow n^2 - 7n + 6n - 42 = 0$$

$$n(n-7) + 6(n-7) = 0 \Rightarrow (n+6)(n-7) = 0$$

$$n+6=0 \Rightarrow \boxed{n=-6 \notin \mathbb{N}}$$

$$n-7=0 \Rightarrow \boxed{n=7}$$

$$(ii) {}^nC_3 = 4 \Rightarrow \frac{n!}{3!(n-3)!} = 4 \Rightarrow \frac{n(n-1)(n-2)(n-3)!}{6(n-3)!} = 4$$

$$n(n-1)(n-2) = 24 \Rightarrow n(n^2 - 3n + 2) = 24$$

$$n^3 - 3n^2 + 2n - 24 = 0$$

$$n^3 - 4n^2 + n^2 - 4n + 6n - 24 = 0$$

$$n^2(n-4) + n(n-4) + 6(n-4) = 0$$

$$(n-4)(n^2 + n + 6) = 0$$

$$n-4=0 \Rightarrow \boxed{n=4}$$

$$n^2 + n + 6 = 0$$

$$n = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(6)}}{2(1)}$$

$$n = \frac{-1 \pm \sqrt{1-24}}{2}$$

$$\boxed{n = \frac{-1 \pm \sqrt{-23}}{2} \notin \mathbb{N}}$$

$$(iii) {}^nC_{12} = {}^nC_6$$

$$\binom{n}{n-12} = \binom{n}{6}$$

$$n-12=6 \Rightarrow n=6+12 \Rightarrow \boxed{n=18}$$



$$(iv) {}^nC_{15} = \frac{17 \times 16}{2!}$$

$$\frac{n!}{15!(n-15)!} = \frac{17 \times 16(15!)}{2!(15!)}$$

$$\frac{n!}{15!(n-15)!} = \frac{17!}{15!(17-15)!}$$

$$\Rightarrow \boxed{n = 17}$$

$$(v) {}^nC_5 = {}^nC_4$$

$$\binom{n}{n-5} = \binom{n}{4}$$

$$n-5 = 4 \Rightarrow n = 4 + 5 \Rightarrow \boxed{n = 9}$$

$$(vi) {}^nC_{10} = \frac{12 \times 11}{2!}$$

$$\frac{n!}{10!(n-10)!} = \frac{12 \times 11(10!)}{2!(10!)}$$

$$\frac{n!}{10!(n-10)!} = \frac{12!}{10!(12-10)!}$$

$$\Rightarrow \boxed{n = 12}$$

Q.3 Find the values of  $n$  and  $r$ , when:

$$(i) {}^nP_r = 210 \text{ and } {}^nC_r = 35$$

$$(ii) {}^{n-1}C_{r-1} : {}^nC_r : {}^{n+1}C_{r+1} = 3 : 6 : 11$$

SOLUTION:

$$(i) {}^nP_r = 210 \text{ and } {}^nC_r = 35$$

$${}^nP_r = {}^nC_r(r!)$$

$$210 = 35(r!)$$

$$r! = \frac{210}{35}$$

$$\frac{n!}{15!(n-15)!} = \frac{17 \times 16 \times 15!}{2!(15!)}$$

$$n! = (n-1)(n-2) \dots$$

$$n!$$

$$n! = 17!$$

$$n = 17$$

$$\frac{n!}{(n-r)!} = r! \frac{n!}{r!(n-r)!}$$

$$r! = 6$$

$$r! = 3!$$

$$\boxed{r = 3}$$

3 × 2 × 1

$${}^n P_r = 210$$

$$\frac{n!}{(n-3)!} = 210$$

$$\frac{n(n-1)(n-2)(n-3)!}{(n-3)!} = 210$$

$$n(n-1)(n-2) = 210$$

$$n(n^2 - 3n + 2) = 210$$

$$n^3 - 3n^2 + 2n = 210$$

$$n^3 - 3n^2 + 2n - 210 = 0$$

$$n^3 - 7n^2 + 4n^2 - 28n + 30n - 210 = 0$$

$$n^2(n-7) + 4n(n-7) + 30(n-7) = 0$$

$$(n-7)(n^2 + 4n + 30) = 0$$

$$n-7=0 \Rightarrow \boxed{n=7}$$

$$n^2 + 4n + 30 = 0$$

$$n = \frac{-4 \pm \sqrt{(4)^2 - 4(1)(30)}}{2(1)}$$

$$n = \frac{-4 \pm \sqrt{-104}}{2}$$

$$n = \frac{-4 \pm 2\sqrt{-26}}{2}$$

$$n = \frac{2(-2 \pm \sqrt{-26})}{2}$$



$$n = -2 \pm \sqrt{-26} \notin \mathbb{N}$$

$$(ii) {}^{n-1}C_{r-1} : {}^nC_r : {}^{n+1}C_{r+1} = 3:6:11$$

$$\frac{(n-1)!}{\{(n-1)-(r-1)\}!} : \frac{n!}{r!(n-r)!} : \frac{(n+1)!}{\{(n+1)-(r+1)\}!} = 3:6:11$$

$$\frac{(n-1)!}{(r-1)!(n-1-r+1)!} : \frac{n!}{r!(n-r)!} : \frac{(n+1)!}{(r+1)!(n+1-r-1)!} = 3:6:11$$

$$\frac{(n-1)!}{(r-1)!(n-r)!} : \frac{n!}{r!(n-r)!} : \frac{(n+1)!}{(r+1)!(n-r)!} = 3:6:11$$

$$\frac{(n-1)!}{(r-1)!(n-r)!} : \frac{n!}{r!(n-r)!} = \frac{3}{6}$$

$$\frac{(n-1)!}{(r-1)!(n-r)!} : \frac{n(n-1)!}{r(r-1)!(n-r)!} = \frac{1}{2}$$

$$\frac{(n-1)!}{(r-1)!(n-r)!} \left\{ \frac{r(r-1)!(n-r)!}{n(n-1)!} \right\} = \frac{1}{2}$$

$$\frac{r}{n} = \frac{1}{2} \Rightarrow \boxed{n = 2r} \rightarrow (1)$$

$$\frac{n!}{r!(n-r)!} : \frac{(n+1)n!}{(r+1)r!(n-r)!} = \frac{6}{11}$$

$$\frac{n!}{r!(n-r)!} \left\{ \frac{(r+1)r!(n-r)!}{(n+1)n!} \right\} = \frac{6}{11} \Rightarrow$$

$$\frac{r+1}{n+1} = \frac{6}{11}$$

$$11r + 11 = 6(2r + 1)$$

$$11r + 11 = 12r + 6$$

$$11 - 6 = 12r - 11r$$

$$\boxed{r = 5}$$

$$(1) \Rightarrow n = 2(5) \Rightarrow \boxed{n = 10}$$

Q.4 If (i)  ${}^{12}C_r = {}^{12}C_{r+4}$  then find  ${}^6C_r$ .

(ii)  ${}^nC_{12} = {}^nC_{10}$  then find  ${}^{25}C_n$ .

**SOLUTION:**

(i)  ${}^{12}C_r = {}^{12}C_{r+4}$  then find  ${}^6C_r$ .

$$\binom{12}{12-r} = \binom{12}{r+4}$$

$$12-r = r+4$$

$$12-4 = r+r$$

$$2r = 8 \Rightarrow \boxed{r = 4}$$

$$\binom{6}{4} = \frac{6!}{4!(6-4)!} = \frac{6(5)(4!)}{4!(2)!} = \frac{6(5)}{2} = 15$$

(ii)  ${}^nC_{12} = {}^nC_{10}$  then find  ${}^{25}C_n$

$$\binom{n}{n-12} = \binom{n}{10}$$

$$n-12 = 10$$

$$n = 10 + 12 \Rightarrow \boxed{n = 22}$$

$$\binom{25}{22} = \frac{25!}{22!(25-22)!} = \frac{25(24)(23)(22!)}{22!(3)!} = \frac{25(24)(23)}{6} = 2300$$

Q.5 The members of a club are 12 boys and 8 girls. In how many ways,

(i) a committee of 5 members be formed?

(ii) a committee of 4 boys and 3 girls be formed?

**SOLUTION:**

Total boys = 12



$$\text{Total girls} = 8$$

$$\text{Total members} = 12 + 8 = 20$$

$$(i) \text{ Total ways} = \binom{20}{5} = 15504$$

$$(i) \text{ Total ways} = \binom{12}{4} \binom{8}{3} = 27720$$

**Q.6A box contains 7 red balls and 6 black balls. In how many ways can 5 balls be selected such that exactly 3 are red?**

**SOLUTION:**

$$\text{Total red balls} = 7$$

$$\text{Total black balls} = 6$$

5 balls are to be drawn

$$\text{Total ways} = \binom{7}{3} \binom{6}{2} = 525$$

**Q.7How many (a) diagonals and (b) triangles can be drawn in a plane by joining the vertices of the polygon having:**

- (i) 6 sides      (ii) 8 sides      (iii) 10 sides

**SOLUTION:**

$$\text{Total triangles} = \binom{n}{3}$$

$$\text{Total diagonals} = \binom{n}{2} - n$$

(i) 6 sides

$$\text{Total triangles} = \binom{6}{3} = 56$$

$$\text{Total diagonals} = \binom{6}{2} - 6 = 9$$

(ii) 8 sides

$$\text{Total triangles} = \binom{8}{3} = 120$$

$$\text{Total diagonals} = \binom{8}{2} - 8 = 20$$

(iii) 10 sides

$$\text{Total triangles} = \binom{10}{3} = 120$$

$$\text{Total diagonals} = \binom{10}{2} - 10 = 35$$

Q.8 Show that:  ${}^9C_6 + {}^9C_5 = {}^{10}C_6$

SOLUTION:

$$\text{L.H.S} = \binom{9}{6} + \binom{9}{5}$$

$$\text{L.H.S} = \frac{9!}{6!(9-6)!} + \frac{9!}{5!(9-5)!}$$

$$\text{L.H.S} = \frac{9!}{6!(3)!} + \frac{9!}{5!(4)!}$$

$$\text{L.H.S} = \frac{9!}{6(5!)(3!)} + \frac{9!}{5!(4)(3!)}$$

$$\text{L.H.S} = \frac{9!(4) + 9!(6)}{(6!)(4!)}$$

$$\text{L.H.S} = \frac{9!(4+6)}{(6!)(4!)}$$

$$\text{L.H.S} = \frac{9!(10)}{(6!)(10-6)!}$$

$$\text{L.H.S} = \frac{10!}{(6!)(4!)}$$

$$\text{L.H.S} = \binom{10}{6}$$



**Q.9** There are 11 men and 9 women members of a club. How many committees of 8 members can be formed, having

- (i) exactly five men                      (ii) at most five women  
(iii) at least five women

**SOLUTION:**

Total men = 11

Total women = 9

A committee of 8 members is to be formed

**(i) exactly five men**

$$\text{Total ways} = {}^{11}C_5 {}^9C_3 = 38808$$

**(ii) at most five women**

$$\begin{aligned} \text{Total ways} &= {}^{11}C_3 {}^9C_5 + {}^{11}C_4 {}^9C_4 + {}^{11}C_5 {}^9C_3 + {}^{11}C_6 {}^9C_2 \\ &+ {}^{11}C_7 {}^9C_1 + {}^{11}C_8 {}^9C_0 \\ &= 120945 \end{aligned}$$

**(iii) at least five women**

$$\begin{aligned} \text{Total ways} &= {}^{11}C_3 {}^9C_5 + {}^{11}C_2 {}^9C_6 + {}^{11}C_1 {}^9C_7 + {}^{11}C_0 {}^9C_8 \\ &= 25815 \end{aligned}$$

**Q.10** Find the number of combinations of the letters of the word QUESTION taken 5 at a time.

**SOLUTION:**

Total letters = 8

5 letters are selected at a time

Total ways =  $\binom{8}{5} = 56$

**Q.11** How many words can be formed by 3 vowels and 4 consonants out of 5 vowels and 7 consonants.

**SOLUTION:**

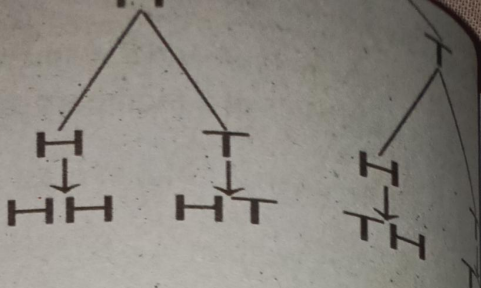
Total vowels = 3

Total consonants = 4

Total letters =  $3 + 4 = 7$

A word of 5 vowels and 7 consonants is to be formed

Total ways =  $\binom{5}{3} \binom{7}{4} (7!) = 1764000$



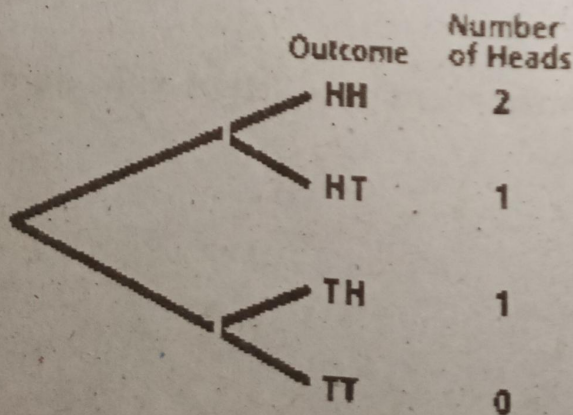
### EXERCISE 6.4

**Q.1** Two fair coins are tossed. Find the probability of getting:

- (i) same faces
- (ii) all heads
- (iii) at most one head
- (iv) at least two tail
- (v) at least one tail

**SOLUTION.**

A coin is tossed two times





$$S = \{HH, HT, TH, TT\}$$

$$O(S) = 4$$

(i) same faces

$$A = \{HH, TT\}$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{2}{4} = \frac{1}{2}$$

(ii) all heads

$$B = \{HH\}$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{1}{4}$$

(iii) at most one head

$$C = \{HT, TH, TT\}$$

$$P(C) = \frac{O(C)}{O(S)} = \frac{3}{4}$$

(iv) at least two tail

$$D = \{TT\}$$

$$P(D) = \frac{O(D)}{O(S)} = \frac{1}{4}$$

(v) at least one tail

$$E = \{HT, TH, TT\}$$

$$P(E) = \frac{O(E)}{O(S)} = \frac{3}{4}$$

**Q.2** Two dice, one red and the other green, are rolled simultaneously. The numbers of dots on the top are added. Find the probability of getting a sum of: (i) 8 (ii) 10 (ii) 12

**SOLUTION:**

Two dice are rolled once

$$S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$O(S) = 6 \times 6 = 36$$

(i) 8

$$A = \{(2,6), (3,5), (5,3), (4,4), (6,2)\}$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{5}{36}$$

(ii) 10

$$B = \{(4,6), (6,4), (5,5)\}$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{3}{36} = \frac{1}{12}$$

(ii) 12

$$C = \{(6,6)\}$$

$$P(C) = \frac{O(C)}{O(S)} = \frac{1}{36}$$

**Q.3** A bag contains 40 balls out of which 5 are green, 15 are red and the remaining are black. A ball is drawn out of the bag. Find the probability of getting:

(i) the ball is green      (ii) the ball is black

(iii) the ball is not green

**SOLUTION:**

Total green balls = 5

Total red balls = 15

Total black balls =  $40 - 20 = 20$

A ball is drawn



(i) the ball is green

$$P(\text{green}) = \frac{5}{40} = \frac{1}{8}$$

(ii) the ball is black

$$P(\text{black}) = \frac{20}{40} = \frac{1}{2}$$

(iii) the ball is not green

$$P(\text{not green}) = 1 - P(\text{green}) = 1 - \frac{1}{8} = \frac{7}{8}$$

**Q.4A** card is drawn from a well shuffled deck of cards, find the probability of: (i) getting a king (ii) getting a club (iii) getting a face card

**SOLUTION:**

(i) getting a king

Total king = 4

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13}$$

(ii) getting a club

Total club = 13

$$P(\text{club}) = \frac{13}{52} = \frac{1}{4}$$

(iii) getting a face card

Total face card = 12

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$$

**Q.5A** dice is rolled. Find the probability using venn diagram of getting:

(i) an even number      (ii) a number greater than 4

(ii) a number which is even and greater than 4

**SOLUTION:**

Total outcomes =  $\{1, 2, 3, 4, 5, 6\}$

**(i) an even number**

Total even number =  $\{2, 4, 6\}$

$$P(\text{even number}) = \frac{3}{6} = \frac{1}{2}$$

**(ii) a number greater than 4**

Total favorable outcomes =  $\{5, 6\}$

$$P(x > 4) = \frac{2}{6} = \frac{1}{3}$$

**(ii) a number which is even and greater than 4**

Total favorable outcomes =  $\{6\}$

$$P(\text{even and greater than 4}) = \frac{1}{6}$$

**Q.6 In a three-child family, by using tree diagram. Find the probability of having:**

(i) three boys      (ii) exactly two boys      (iii) at least one girl

**SOLUTION:**

$$S = \{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG\}$$

$$O(S) = 2^3 = 8$$



(i) three boys

$$A = \{BBB\}$$

$$O(A) = 1$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{1}{8}$$

(ii) exactly two boys

$$B = \{BBG, GBB, BGB\}$$

$$O(B) = 3$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{3}{8}$$

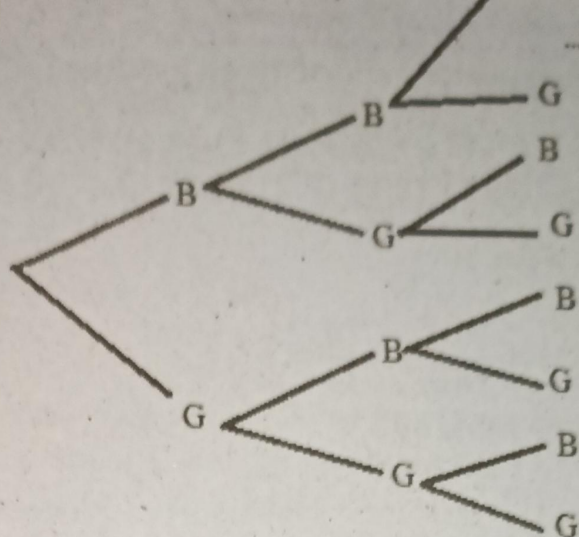
(iii) at least one girl

$$C = \{BBG, BGB, BGG, GBB, GBG, GGB, GGG\}$$

$$O(C) = 7$$

$$P(C) = \frac{O(C)}{O(S)} = \frac{7}{8}$$

$$\text{OR } P(C) = \frac{O(C)}{O(S)} = \frac{8-1}{8} = \frac{7}{8}$$



**Q.7** In a single throw of two fair dice, find the probability that the product of the number on the dice is:

- (i) between 2 and 10 (both inclusive)      (ii) divisible by 5

**SOLUTION:**

Two dice are rolled once

$$S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$O(S) = 6 \times 6 = 36$$

(i) between 2 and 10 (both inclusive)

$$A = \{ (1,2), (2,1), (1,3), (3,1), (1,4), (4,1), (1,5), (5,1), (1,6), \\ ((6,1), (6,1), (2,2), (2,3), (2,4), (2,5), (3,2), (3,3), (4,2), (5,2)) \}$$

$$O(A) = 18$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{18}{36} = \frac{1}{2}$$

(ii) divisible by 5

$$B = \{ (1,5), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), \\ (2,5), (3,5), (4,5), (6,5) \}$$

$$O(B) = 11$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{11}{36}$$

**Q.8A** marble is drawn at random from a box containing 20 red, 10 white, 25 orange and 15 blue marbles. Find the probability that it is: (i) orange or red (ii) not blue or red (iii) red, white or blue

**SOLUTION:**

Total red marbles = 20

Total white marbles = 10

Total orange marbles = 25

Total blue marbles = 15

Total marbles =  $20 + 10 + 25 + 15 = 70$

(i) orange or red

$$P(\text{orange or red}) = \frac{25 + 20}{70} = \frac{45}{70} = \frac{9}{14}$$

(ii) not blue or red



$$P(\text{not blue or red}) = \frac{70 - (15 + 20)}{70} = \frac{70 - 35}{70} = \frac{35}{70} = \frac{1}{2}$$

(iii) red, white or blue

$$P(\text{red, white or blue}) = \frac{20 + 10 + 15}{70} = \frac{45}{70} = \frac{9}{14}$$

**Q.9** The king, queen and jack of clubs are removed from a deck of 52 playing cards and then shuffled. A card is drawn from the remaining cards. Find the probability of getting:

(i) heart  
red card

(ii) a queen

(iii) a club

(iv) 9 of

**SOLUTION:**

Total cards after removing king, queen and jack =  $52 - 3 = 49$

(i) heart

$$P(\text{heart}) = \frac{13}{49}$$

(ii) a queen

$$P(\text{queen}) = \frac{3}{49}$$

(iii) a club

$$P(\text{club}) = \frac{10}{49}$$

(iv) 9 of red card

$$P(9 \text{ of red card}) = \frac{2}{49}$$

**Q.10** A pair of fair dice is thrown. If the two numbers appearing are different, find the probability that (i) the sum is 10 (ii) the sum of six or less.

**SOLUTION:**

## A Help Book of Mathematics XI

Two dice are rolled once

$$S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$O(S) = 6 \times 6 = 36$$

(i) the sum is 10

$$A = \{(4,6), (6,4)\}$$

$$O(A) = 2$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{2}{36} = \frac{1}{18}$$

(ii) the sum of six or less

$$B = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (2,1), (2,2), (2,3), \\ (2,4), (3,1), (3,2), (3,3), (4,1), (4,2), (5,1) \end{array} \right\}$$

$$O(B) = 12$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{12}{36} = \frac{1}{3}$$

**Q.11** Given that  $P(A) = 0.3$ ,  $P(B) = 0.7$ ,  $P(A \cap B) = 0.21$  then find: (i)  $P(A/B)$  (ii)  $P(B/A)$

**SOLUTION:**

$$P(A) = 0.3, P(B) = 0.7, P(A \cap B) = 0.21$$

$$(i) P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0.21}{0.7} = 0.3$$

(ii)  $P(B/A)$

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{0.21}{0.3} = 0.7$$



**Q.12** If a card is selected at random from a deck of 52 playing cards, what is the probability that the card is a club or a face or both?

**SOLUTION:**

$$\text{Total cards} = 52$$

$$\text{Total club} = 13$$

$$\text{Total face card other than club} = 12 - 3 = 9$$

$$P(\text{cards or club}) = P(A \cup B)$$

$$= P(A) + P(B)$$

$$= \frac{13}{52} + \frac{9}{52} = \frac{11}{26}$$

**Q.13** From two events A and B,  $P(A) = 0.5$ ,  $P(B) = 0.2$ ,  $P(A \cup B) = 0.4$  then find  $P(A \cap B)$ .

**SOLUTION:**

$$P(A) = 0.5, P(B) = 0.2, P(A \cup B) = 0.4$$

$$P(A \cap B) + P(A \cup B) = P(A) + P(B)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$P(A \cap B) = 0.5 + 0.2 - 0.4$$

$$P(A \cap B) = 0.3$$

**Q.14** A natural number is chosen out of first 35 natural numbers. What is the probability that the chosen number is divisible by 8 or 9?

**SOLUTION:**

$$S = \{1, 2, 3, \dots, 35\}$$

$$A = \{8, 16, 24, 32\}$$

$$B = \{9, 18, 27\}$$

∴ A and B are disjoint events

$$P(A \text{ or } B) = P(A \cap B) = P(A) + P(B)$$

$$P(A \text{ or } B) = \frac{3}{35} + \frac{4}{35} = \frac{7}{35}$$

$$P(A \cap B) = \frac{1}{5}$$

**Q.15** A bag contains 15 black, 25 red and 10 white balls. A ball is drawn at random. Find the probability that it is either red or white?

**SOLUTION:**

Total black balls = 15

Total red balls = 25

Total white balls = 10

Total balls = 15 + 25 + 10 = 50

$$P(\text{red or white}) = \frac{25 + 10}{50} = \frac{35}{50} = \frac{7}{10}$$

**Q.16** Two events A and B are such that  $P(A) = \frac{1}{4}$ ,  $P(A/B) = \frac{1}{2}$ ,  $P(B/A) = \frac{2}{3}$ . Find  $P(A \cap B)$  and  $P(B)$ .

**SOLUTION:**

$$P(A) = \frac{1}{4}, P\left(\frac{A}{B}\right) = \frac{1}{2}, P(B/A) = \frac{2}{3}$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$\frac{2}{3} = \frac{P(A \cap B)}{\frac{1}{4}} \Rightarrow \frac{2}{3} \left(\frac{1}{4}\right) = P(A \cap B) \Rightarrow P(A \cap B) = \frac{1}{6}$$



$$P(A/B) = \frac{P(A \cap B)}{P(B)} \Rightarrow \frac{1}{2} = \frac{\frac{1}{6}}{P(B)} \Rightarrow P(B) = 2 \left( \frac{1}{6} \right) \Rightarrow \boxed{P(B) = \frac{1}{3}}$$

Q.17A fair die is thrown twice. Find the probability that an even number of dots appear in first and the number of dots in the second throw is less than 4?

SOLUTION:

For a single throw of a die,  $S = \{1, 2, 3, 4, 5, 6\}$

$$A = \{2, 4, 6\}, B = \{1, 2, 3\}$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{3}{6} = \frac{1}{2}$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{3}{6} = \frac{1}{2}$$

Since A and B are independent events

$$P(A \cap B) = P(A)P(B) = \left( \frac{1}{2} \right) \left( \frac{1}{2} \right) = \frac{1}{4}$$

Q.18Three missiles are fired at a target. If the probability of hitting the target is 0.5, 0.3 and 0.6 respectively, and if the missiles are fired independently, what is the probability that all the missiles hit the target?

SOLUTION:

$$P(A) = (0.5), P(B) = (0.3), P(C) = (0.6)$$

Since A, B and C are independent events

$$P(A \cap B \cap C) = P(A)P(B)P(C) = (0.5)(0.3)(0.6) = 0.09$$

### REVIEW EXERCISE 6

1. Select the correct answer. ✓

(i) If  $n = 0$ , then  $\frac{(n+1)!}{n!} = \underline{\hspace{2cm}}$

(a) 0

(b) 1

(c)  $n$

(d)  $\infty$

(ii) Probability of getting 7 in throwing a dice is:

(a) 0

(b) 1

(c) -1

(d) not defined

(iii) The factorial form of 12. 11. 10 is:

(a)  $\frac{12!}{9!}$

(b) 12!

(c)  $\left(\frac{12}{9}\right)!$

(d)  $(12!)(9!)$

(iv) If two independent events A and B occur in  $p$  and  $q$  ways respectively, then number of ways that both events can occur is:

(a)  $p + q$  ways

(b)  $pq$  ways

(c)  $(pq)^r$  ways

(d)  $rp + qr$  ways

(v) An arrangement of  $n$  objects according to some definite order is called:

(a) combination

(b) permutation

(c) factorial

(d) none of these

(vi) An arrangement of  $n$  objects without any order is called:

(a) combination

(b) permutation

(c) factorial

(d) none of these

(vii) The number of permutations of the letter of the word COMMITTEE is:

(a)  $\binom{9}{2, 2, 2}$

(b)  $\binom{6}{1, 2, 2}$



(c)  $\binom{9}{2,2,1}$

(d)  $\binom{9}{2,3,1}$

(viii) 8. 7. 6 is equal to:

(a)  ${}^8P_3$

(b)  ${}^8C_3$

(c)  ${}^8P_5$

(d)  ${}^8C_5$

(ix) If  $r = n$ , then  ${}^nP_r$  is equal to:

(a)  $r!$

(b)  $(n - r)!$

(c) 1

(d) 0

(x) The number of ways that a necklace of  $n$  beads of different colours be made is:

(a)  $(n - 1)!$

(b)  $\frac{n!}{2}$

(c)  $\frac{n! - 1}{2}$

(d)  $\frac{(n-1)!}{2}$

(xi) Any subset of a sample space is called:

(a) sample space

(b) an event

(c) a trial

(d) random variable

(xii) For two events A and B if  $A \cap B = \phi$ , then events A and B are called:

(a) mutually exclusive

(b) not mutually exclusive

(c) overlapping

(d) dependent events

(xiii) When a dice is rolled and coin is tossed, all possible outcome are:

(a) 6

(b) 12

(c) 18

(d) 24

(xiv) If two events A and B have equal chance of occurrence, then the events are:

- (a) equally likely (b) not equally likely  
(c) dependent (d) not mutually exclusive

(xv) If E be an event of a sample space S, then:

- (a)  $P(E) = \frac{n(S)}{n(E)}$  (b)  $0 \leq P(E) \leq 1$   
(c)  $0 < P(E) < 1$  (d) all of these

(xvi) The probability of getting the tail in a single toss of a coin is:

- (a)  $\frac{1}{3}$  (b)  $\frac{2}{3}$   
(c)  $\frac{1}{2}$  (d)  $\frac{1}{4}$

(xvii) Three dice are rolled simultaneously, then  $n(S)$  is equal to:

- (a) 36 (b) 18  
(c) 216 (d) 6

(xviii) Two teams A and B playing a match, the probability that team A does not lose is:

- (a)  $\frac{1}{3}$  (b)  $\frac{2}{3}$   
(c)  $\frac{1}{2}$  (d) 0

(xix) If  ${}^nC_6 = {}^nC_{12}$ , then n equals:

- (a) 18 (b) 12  
(c) 6 (d) 20

Q.2 Evaluate the following:



$$(i) \frac{{}^6C_3 \times {}^4C_3}{{}^{10}C_4} \quad (ii) \frac{{}^6P_3 \times {}^4P_3}{{}^{10}P_4} \quad (iii) \sum_{r=1}^3 {}^3C_r$$

**SOLUTION:**

$$(i) \frac{{}^6C_3 \times {}^4C_3}{{}^{10}C_4}$$

$$= \left\{ \frac{6!}{3!(6-3)!} \right\} \left\{ \frac{4!}{3!(4-3)!} \right\} \div \left\{ \frac{10!}{4!(10-4)!} \right\}$$

$$= \left\{ \frac{6!}{3!(3)!} \right\} \left\{ \frac{4!}{3!(1)!} \right\} \div \left\{ \frac{10!}{4!(6)!} \right\}$$

$$= \left\{ \frac{6(5)(4)(3!)}{3!(3)!} \right\} \left\{ \frac{4(3!)}{3!} \right\} \div \left\{ \frac{10(9)(8)(7)(6!)}{4!(6)!} \right\}$$

$$= \left\{ \frac{6(5)(4)}{3!} \right\} \{4\} \div \left\{ \frac{10(9)(8)(7)}{4!} \right\}$$

$$= \frac{80}{210}$$

$$= \frac{8}{21}$$

$$(ii) \frac{{}^6P_3 \times {}^4P_3}{{}^{10}P_4}$$

$$= \left\{ \frac{6!}{(6-3)!} \right\} \left\{ \frac{4!}{(4-3)!} \right\} \div \left\{ \frac{10!}{(10-4)!} \right\}$$

$$= \left\{ \frac{6!}{(3)!} \right\} \left\{ \frac{4!}{(1)!} \right\} \div \left\{ \frac{10!}{(6)!} \right\}$$

$$= \left\{ \frac{6(5)(4)(3!)}{(3)!} \right\} \{24\} \div \left\{ \frac{10(9)(8)(7)(6!)}{(6)!} \right\}$$

$$= \{120\} \{24\} \div \{5040\}$$

$$= \frac{4}{7}$$

$$(iii) \sum_{r=1}^3 {}^3C_r$$

$$= \binom{3}{1} + \binom{3}{2} + \binom{3}{3}$$

$$= 3 + 3 + 1$$

$$= 7$$

**Q.3** There are seven seats available in a compartment. In how many ways can seven persons be seated?

**SOLUTION:**

$$\text{Total seats} = 7$$

$$\text{Total persons} = 7$$

$$\text{Total ways} = {}^7P_7 = 5040$$

**Q.4** A room has 3 lamps. From a collection of 10 light bulbs of which 6 are no good, a person selects 3 at random and puts them in the sockets. What is the probability that he will have light?

**SOLUTION:**

$$\text{Total bulbs} = 10$$

$$\text{Total defective bulbs} = 6$$

$$\text{Total good bulbs} = 10 - 6 = 4$$

3 bulbs are selected at random

$$\begin{aligned} \text{Total ways} &= {}^4C_1 {}^6C_2 + {}^4C_2 {}^6C_1 + {}^4C_3 {}^6C_0 \\ &= (4)(15) + (6)(6) + (4)(1) \\ &= 100 \end{aligned}$$

**Q.5** If two dice are thrown simultaneously, what is the probability of obtaining a sum of 7 or a sum of 11?

**SOLUTION:**

Two dice are rolled once



$$S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

$$O(S) = 6 \times 6 = 36$$

$$A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1), (5,6), (6,5)\}$$

$$O(A) = 11$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{11}{36}$$

**Q.6a** bag contains 8 white, 10 black and 12 red balls. 3 balls are drawn at from the bag. What is the probability that the first ball is white, the second ball is black, and the third ball is red, when every time the ball is replaced?

**SOLUTION:**

Total white balls = 8

Total black balls = 10

Total red balls = 12

Total balls =  $8 + 10 + 12 = 30$

3 balls are selected after replacement i.e., events are independent

$$P(A \cap B \cap C) = P(A)P(B)P(C) = \left(\frac{8}{30}\right)\left(\frac{10}{30}\right)\left(\frac{12}{30}\right) = \frac{8}{225}$$

**Q.7** In how many ways can a football team of 11 players be selected out of 15 players? How many of them will include a particular player?

**SOLUTION:**

Total players = 15

11 players are to be selected

(i) Total ways =  $\binom{15}{11} = 1365$

(ii) Total ways =  $\binom{1}{1} \binom{14}{10} = 1001$